

Multi-Curve Convexity

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Abstract

In this paper we analyse the impact of tenor basis in a multi-curve environment on the pricing of CMS instruments. CMS derivatives may be priced by means of a replication approach. Doing so the derivative is translated into a portfolio of Vanilla swaptions. On the one side multi-curve pricing enters into this approach by valuing the replicating vanilla swaptions. On the other side the basis between forward and discount curves also impacts the specification of the replication portfolio.

The specification of the replication portfolio is driven by the definition of an annuity mapping function. That function links the quotient of the discount factor for the pay date and the underlying swap rate annuity to the underlying swap rate. In this study we investigate the impact of the multi-curve setting on the specification of the annuity mapping function. As a result we find that the impact of multi-curve setting on the annuity mapping function is small compared to the impact of the multi-curve setting on the underlying Vanilla swaption prices.

Keywords: CMS replication, multi-curve pricing, tenor and funding basis.

1 Introduction and Notation

Pricing of constant maturity swap (CMS) derivatives is often realised by means of the replication approach elaborated e.g. in [Hag03]. This approach links CMS payoffs to a portfolio of Vanilla options for which prices are (in principle) observable in the market or can be evaluated by the Vanilla pricing model of choice.

An integral quantity for CMS pricing (in the annuity measure) is the quotient of the discount factor for the pay date and the underlying swap rate annuity (as observed at the fixing date). Application of the replication approach requires representing this quotient as a function of the underlying swap rate itself. In [Hag03] this representation is considered a *yield curve model*. A more formal approach is derived in [AP10, Sec. 16.6.2] which considers that quotient as an *annuity mapping function* that represents a conditional expectation. For this study we will resort to the representation of the annuity mapping function as elaborated in [AP10, Sec. 16.6.2].

Several candidates of annuity mapping functions are analysed e.g. in [Hag03] and [AP10, Sec. 16.5, 16.6]. A further contribution for deriving an annuity mapping function is presented in [CP12]. In that study the annuity mapping function is fully implied from all volatilities and correlations involved. Thus the risk w.r.t. these inputs can be estimated. A relevant sub-class of annuity mapping functions are affine functions in the swap rate (or linearisations of the non-linear functions cited above). These instances improve analytical tractability of the overall model. They are usually denoted as linear terminal swap rate (TSR) models.

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If a full term structure model is applied the annuity mapping function is specified by the model dynamics of discount factors (or zero bonds) in a consistent and arbitrage-free way. However, usually the annuity mapping function is specified exogenously. Thus no-arbitrage and consistency conditions need to be imposed in order to derive a meaningful model. The conditions in question are formulated, e.g. in [CP12, Eq. (6) - (8)].

Another relevant aspect is the incorporation of the basis between discount curves and forward Libor rates. This basis usually originates from the tenor basis between Libor and OIS. If the derivative is not cash collateralised (in the coupon currency) further funding basis spreads need to be taken into account. Obviously, the basis needs to be captured when pricing the replicating portfolio of Vanilla options. But the basis also enters the construction of the replication portfolio itself.

In this study we analyse the impact of the basis between discounting and forward curves on the convexity adjustment of CMS payoffs. We use a basis model based on continuous compounded forward rate spreads as proposed e.g. in [AP10, Sec. 15.5] and further analysed in [SM15]. This representation allows taking into account tenor and funding spreads for Libor and swap rates in a consistent fashion. As widely adapted in the industry we assume that this basis is deterministic and can be estimated from today's yield curves. Given that our focus is on quantifying the impact of the basis we use a linear TSR model with a fairly simple parametrisation.

The impact of Libor versus OIS basis is also considered in [CP12]. Moreover, in [Pit10] the impact of a stochastic funding basis is characterised by means of an annuity mapping function. In [CP12] the spread is considered between Libor and OIS-based (simple compounded) forward swap rates. With this basis spread representation a linear TSR model will only be impacted if the basis is considered stochastic. Moreover, as discussed in [SM15] a deterministic spread assumption based on simple compounded (or market) rates may prove problematic in non-OIS funding situations.

In this study we take an alternative view and define basis spreads related to individual forward Libor rates. Consequently, the spread between Libor and OIS-based swap rates is derived from the individual Libor rates spreads. This gives a consistent basis spread representation across Libor and swap rates.

The remainder of that paper is organised as follows: We introduce basic notations in Sec. 1.1 and specify our basis spread model in Sec. 1.2. In Sec. 2 we analyse the CMS pricing via a terminal swap rate model using an annuity mapping function in the presence of separate curves for forward rate and discount factor evaluation. Finally, in Sec. 3 we apply our model to the pricing of non-standard Libor and CMS payoffs. Moreover, we quantify the impact of the tenor basis on the convexity adjustment.

1.1 Multi-Curve Setup

We consider schedule dates $T_0, T_1, \dots, T_N$ for a floating leg paying Libor coupons. The Libor rate for a time period $[T_{i-1}, T_i]$ is denoted as $L(T_{i-1}, T_i)$. The time period or tenor usually corresponds to three or six months and the corresponding year fraction is τ_i . Corresponding forward Libor rates observed at t are defined as

$$L(t; T_{i-1}, T_i) = \mathbb{E}^{T_i} [L(T_{i-1}, T_i) \mid \mathcal{F}(t)].$$

Here, $\mathbb{E}^{T_i} [\cdot]$ denotes the expectation under the T_i terminal measure.

Derivative discounting is assumed to be based on overnight or other funding curves. $P(t, T)$ denotes the discount factor at observation time t with maturity T .

Vanilla rates derivatives are caps and swaptions. Caps are given as a sum of caplets where each caplet is a call option on the forward Libor rate with strike K accrued with year fraction τ_i and with payoff paid at T_i . Today's price of a caplet is

$$\text{Caplet}_i(t) = P(t, T_i) \cdot \tau_i \cdot \mathbb{E}^{T_i} [(L(T_{i-1}, T_i) - K)^+ \mid \mathcal{F}(t)].$$

A (physically settled) payer swaption is an option to enter into a fixed versus floating swap. The fixed leg schedule is determined by $\bar{T}_0, \bar{T}_1, \dots, \bar{T}_M$ with year fractions $\bar{\tau}_1, \dots, \bar{\tau}_M$. The present value of

the underlying swap with fixed rate K at exercise time T is

$$Swap(t) = \sum_{i=1}^N L(t, T_{i-1}, T_i) \tau_i P(t, T_i) - K \cdot \sum_{j=1}^M \bar{\tau}_j P(t, \bar{T}_j).$$

Re-writing the swap in terms of annuity

$$An(t) = \sum_{j=1}^M \bar{\tau}_j P(t, \bar{T}_j)$$

and swap rate

$$S(t) = \frac{\sum_{i=1}^N L(t, T_{i-1}, T_i) \tau_i P(t, T_i)}{An(t)}$$

allows representing the swaption value as

$$Swaption(t) = An(t) \cdot \mathbb{E}^{An} \left[(S(T) - K)^+ \mid \mathcal{F}(t) \right].$$

In this representation $\mathbb{E}^{An}[\cdot]$ is the expectation in the corresponding annuity measure.

Both caplets and swaptions are usually priced by means of single rate models that describe the dynamics of the Libor forward rate or the swap rate in their respective martingale measure. The popular SABR model (or variants thereof) has emerged as quasi industry standard for pricing caplets and swaptions.

Also it is common to represent prices of caplets and swaptions in terms of log-normal or normal implied volatilities based on the Black and Bachelier formula. For details, see e.g. [BP12]. Since modelling negative rates has become increasingly important in recent years we base parts of our analysis on normal volatilities instead of log-normal volatilities.

1.2 Deterministic Tenor Basis Model

So far we did not specify any relation between forward Libor rates $L(t; T_{i-1}, T_i)$ and discount factors $P(t, T)$. This is not explicitly required as long as linear products or standard-tenor options are concerned. For our purpose however we need to specify and model the basis between discount curve and forward Libor rates.

The deterministic modelling of tenor and funding basis is analysed e.g. in [SM15]. In particular tenor basis may either be modelled as spread on simple compounded rates or continuously compounded rates. As presented in [SM15] it appears more advantageous to use continuously compounded rates spreads in the context of various funding assumptions. Thus we use rates spreads based on continuous compounding for the further analysis. Nevertheless, the results can easily be adapted to rates spreads based on simple compounding. For details see Appendix A.3.

Instantaneous forward rates of the discount curve are given by $f(t, T) = -\partial \log(P(t, T)) / \partial T$. We assume that forward Libor rates are parametrised by a forward curve with (pseudo) discount factors $P^{fw}(t, T)$ and the relation

$$L(t; T_{i-1}, T_i) = \left[\frac{P^{fw}(t, T_{i-1})}{P^{fw}(t, T_i)} - 1 \right] \frac{1}{\tau_i}.$$

This allows us defining the continuous forward rates of the Libor forward curve as

$$f^{fw}(t, T) = -\frac{\partial \log(P^{fw}(t, T))}{\partial T}.$$

The spread between continuous compounded Libor forward rates and discount rates is

$$b(t, T) = f^{fw}(t, T) - f(t, T).$$

Our key modelling assumption is that the spread $b^{fw}(t, T)$ is deterministic and independent of the observation time t . That is we require that we can deduce the term structure of future tenor basis from today's yield curves.

With

$$D(T_{i-1}, T_i) = e^{\int_{T_{i-1}}^{T_i} b(\cdot, u) du}$$

we may now write

$$1 + \tau_i \cdot L(t; T_{i-1}, T_i) = \frac{P^{fw}(t, T_{i-1})}{P^{fw}(t, T_i)} = D(T_{i-1}, T_i) \cdot \frac{P(t, T_{i-1})}{P(t, T_i)} \quad (1)$$

The condition in (1) essentially connects forward Libor rates and discount factors by our deterministic tenor spread assumption. Moreover, we can use the deterministic spread assumption to write forward swap rates solely based on discount factors as stochastic quantities.

We simplify notation by setting $L_i(t) = L(t, T_{i-1}, T_i)$ and $D_i = D(T_{i-1}, T_i)$ and get

$$L_i(t) \cdot \tau_i \cdot P(t, T_i) = D_i \cdot P(t, T_{i-1}) - P(t, T_i)$$

and

$$\begin{aligned} S(t) &= \frac{\sum_{i=1}^N [D_i \cdot P(t, T_{i-1}) - P(t, T_i)]}{An(t)} \\ &= \frac{P(t, T_0) - P(t, T_N) + \sum_{i=1}^N [D_i - 1] \cdot P(t, T_{i-1})}{An(t)} \end{aligned}$$

Note that the above swap rate formula simplifies to the classical (single-curve setting) formula $S(t) = [P(t, T_0) - P(t, T_N)] / An(t)$ if tenor basis vanishes and thus $D_i = 1$. For briefly we simplify notation further to

$$S(t) = \frac{\sum_{i=0}^N w_i P(t, T_i)}{An(t)}. \quad (2)$$

with weights w_i as

$$\begin{aligned} w_0 &= D_1, \\ w_i &= [D_{i+1} - 1] \text{ for } i = 1, \dots, N-1 \text{ and} \\ w_N &= -1. \end{aligned} \quad (3)$$

2 Terminal Swap Rate Model

Terminal swap rate (TSR) models may be applied to price *approximately single-rate* derivatives as presented in [AP10, Sec. 16.3]. We consider a payoff function $h(S(T))$ depending on the fixing of a swap rate at exercise date T . The payoff is paid at $T_p \geq T$. Typical payoffs are CMS swap coupons with $h(S(T)) = S(T)$ or CMS caplets and floorlets with $h(S(T)) = [S(T) - K]^+$ and $h(S(T)) = [K - S(T)]^+$ respectively.

Today's value of the derivative payoff is then

$$V(t) = An(t) \cdot E^{An} \left[\frac{P(T, T_p)}{An(T)} \cdot h(S(T)) \right].$$

A common equivalent cash-flow based representation is

$$V(t) = P(t, T_p) \cdot \{E^{An} [h(S(T))] + CA(t)\}$$

with the convexity adjustent $CA(t)$ defined as

$$CA(t) = E^{An} \left[\left[\frac{An(t)}{P(t, T_p)} \cdot \frac{P(T, T_p)}{An(T)} - 1 \right] \cdot h(S(T)) \right]. \quad (4)$$

In general, valuation of the above formulas requires knowledge of the dynamics of the swap rate as well as the dynamics of discount factors. This information might be available e.g. in a full term structure model. However, in that case we are likely to loose either flexibility regarding model dynamics or (semi-analytical) tractability of the model.

In the next section we analyze TSR models that take into account the tenor basis. And we will estimate the impact arising from the incorporation of the multi-curve framework.

2.1 Annuity Mapping Function

In practice it turns out that the dynamics of swap rate and discount factors are closely linked. This motivates expressing the quotient of discount factors $P(T, T_p)$ and annuity $An(T)$ also in terms of the swap rate $S(T)$. For this purpose consider the iterated expectation

$$E^{An} \left[E^{An} \left[\frac{P(T, T_p)}{An(T)} \cdot h(S(T)) \mid S(T) = s \right] \right] = E^{An} \left[E^{An} \left[\frac{P(T, T_p)}{An(T)} \mid S(T) = s \right] \cdot h(S(T)) \right].$$

According to [AP10, Sec. 16.6.4] we define an annuity mapping function as

$$\alpha(s, T_p) = E^{An} \left[\frac{P(T, T_p)}{An(T)} \mid S(T) = s \right].$$

Note that by construction $\alpha(s, T_p)$ is a deterministic function in s . If we find a functional relation for $\alpha(s, T_p)$ then we may write the payoff as

$$V(t) = An(t) \cdot E^{An} [\alpha(S(T), T_p) \cdot h(S(T))].$$

The latter may then be solved solely by knowing the dynamics of the swap rate $S(T)$.

To get an impression of the shape and properties of an annuity mapping function we give an example in Figure 1 and Figure 2. Here we consider a 10y x 10y swap rate based on 6M Euribor and our deterministic tenor spread model from Sec. 1.2. The annuity mapping function is implied from a Hull White model calibrated to EUR market data as of September 2015 with mean reversion marked at 1% and 10% respectively.

We see in Figure 1 that for fixed pay times T_p the annuity mapping function is fairly linear for swap rate values in a reasonable range applicable for replication, say $[-10\%, +10\%]$. Nevertheless there is some convexity in the wings (otherwise discount factors would need to be negative). Also we see in Figure 2 that for fixed swap rates $S(T)$ the annuity mapping function is also more or less linear in T_p . Moreover, the impact of the mean reversion parameter in the Hull White model appears very limited. Though the Hull White model is a fairly simple model (not capturing smile and rates de-correlation) this example demonstrates the general behaviour of an annuity mapping function.

In practical applications the functional form of the annuity mapping function usually does not origin from a full term structure model as in Figure 1 and Figure 2. Hence no-arbitrage and consistency of the resulting model are not given a priori but need to be enforced exogenously. We consider the following model-independent conditios:

1. No-arbitrage condition

$$E^{An} [\alpha(S(T), T_p)] = \frac{P(t, T_p)}{An(t)} \text{ for all } T_p \geq T, \quad (5)$$

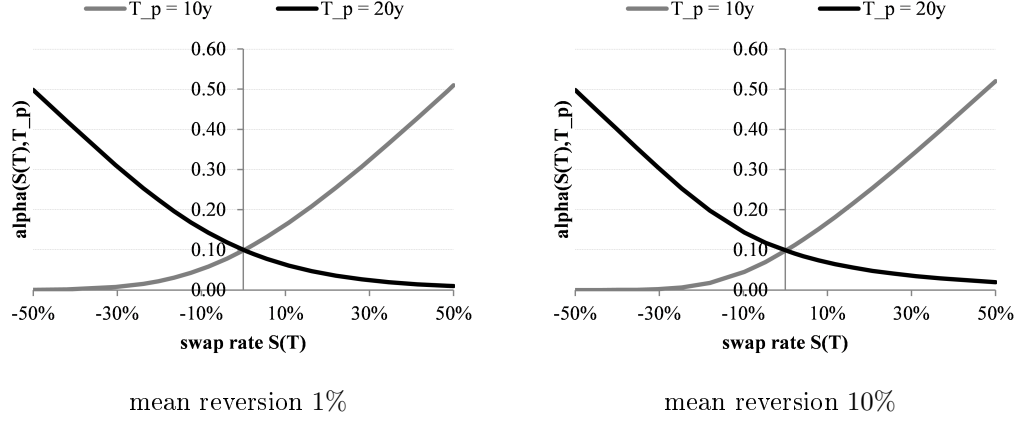


Figure 1: Annuity mapping function $\alpha(\cdot, T_p)$ for a 10y x 10y swap rate in a Hull White model ($T_p \in \{10y, 20y\}$) calibrated to EUR market data as of September 2015 (mean reversion 1% and 10%).

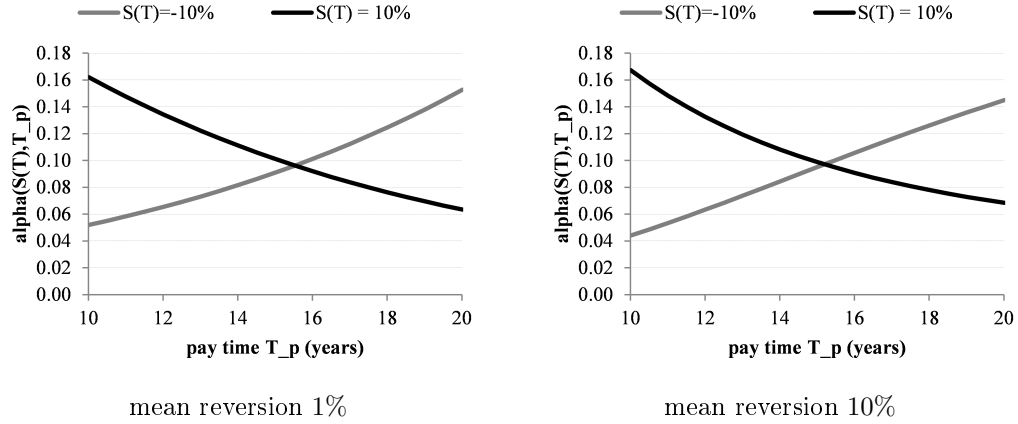


Figure 2: Annuity mapping function $\alpha(S(T), \cdot)$ for a 10y x 10y swap rate in a Hull White model ($S(T) \in \{-10\%, 10\%\}$) calibrated to EUR market data as of September 2015 (mean reversion 1% and 10%).

2. Additivity condition

$$\sum_{j=1}^M \bar{\tau}_j \cdot \alpha(s, \bar{T}_j) = 1 \text{ for all } s, \text{ and} \quad (6)$$

3. Consistency condition

$$\sum_{i=0}^N w_i \cdot \alpha(s, T_i) = s \text{ for all } s. \quad (7)$$

Equivalent representations of the above conditions in a single curve setting are also considered in [CP12]. As pointed out in [AP10] the no-arbitrage condition is by far the most important. But to specify a meaningful TSR model also the additivity and consistency condition should hold.

Interpretation of annuity mapping function conditions. The annuity mapping function $\alpha(s, T_p)$ is given as a two-dimensional function of the underlying state s and the time T_p . The no-arbitrage condition is essentially a condition on the overall (or average) level of the function $\alpha(s, T_p)$ along the s dimension. In contrast, additivity and consistency condition impact $\alpha(s, T_p)$ along the T_p dimension. Here the additivity condition specifies the overall level of $\alpha(\cdot, T_p)$. Unless basis spreads are dominating the overall rates level the consistency condition is approximately $\alpha(s, T_0) - \alpha(s, T_N) = s$. Hence, this condition specifies the average slope of $\alpha(\cdot, T_p)$. Note that the conditions do not specify how $\alpha(\cdot, T_p)$ behaves for individual times T_p .

Multi-curve impact on annuity mapping function. We find that no-arbitrage and additivity condition are not affected by tenor basis. However, the consistency condition does depend on the tenor spreads by the weights w_i coming from (3). If the tenor basis vanishes then the consistency condition involves only $\alpha(s, T_0)$ and $\alpha(s, T_N)$. With non-trivial basis the values $\alpha(s, T_i)$ of all the floating leg dates T_i are involved.

Given that basis spreads enter the consistency condition through the weights w_i we may conclude that the basis spreads essentially impact the slope of the annuity mapping function $\alpha(\cdot, T_p)$ along the T_p dimension.

2.2 Linear TSR with Basis Spreads

For this study we focus on the linear TSR model. The linear TSR model postulates an affine functional relation for the annuity mapping function. That is we set

$$\alpha(s, T_p) = a(T_p) \cdot s + b(T_p) \quad (8)$$

for suitable time-dependent functions $a(\cdot)$ and $b(\cdot)$. Subsequently we elaborate how to specify the functions $a(\cdot)$ and $b(\cdot)$ such that the basic conditions in Sec 2.1 are satisfied. The affine structure of the annuity mapping function yields a fair amount of analytical tractability

No-arbitrage condition. We have from (5) and (8) that

$$E^{An} [\alpha(S(T), T_p)] = E^{An} [a(T_p) \cdot S(T) + b(T_p)] = a(T_p) \cdot S(t) + b(T_p).$$

Thus the no-arbitrage condition yields the relation

$$b(T_p) = \frac{P(t, T_p)}{An(t)} - a(T_p) \cdot S(t) \text{ for all } T_p \geq T.$$

This gives a direct link between the functions $a(\cdot)$ and $b(\cdot)$. Consequently, if we specify the function $a(\cdot)$ then we may directly infer the function $b(\cdot)$. As a result the no-arbitrage condition allows to re-write the annuity mapping function only in terms of $a(\cdot)$ as

$$\alpha(s, T_p) = a(T_p) \cdot [s - S(t)] + \frac{P(t, T_p)}{An(t)}. \quad (9)$$

Additivity condition. Substituting (9) in (6) yields

$$\begin{aligned} \sum_{j=1}^M \bar{\tau}_j \cdot \alpha(s, \bar{T}_j) &= \sum_{j=1}^M \bar{\tau}_j \cdot \left[a(\bar{T}_j) \cdot [s - S(t)] + \frac{P(t, \bar{T}_j)}{An(t)} \right] \\ &= \left[\sum_{j=1}^M \bar{\tau}_j \cdot a(\bar{T}_j) \right] \cdot [s - S(t)] + 1. \end{aligned}$$

Consequently the additivity condition requires that

$$\sum_{j=1}^M \bar{\tau}_j \cdot a(\bar{T}_j) = 0.$$

Consistency condition. Combining (9) and (7) gives

$$\begin{aligned} \sum_{i=0}^N w_i \cdot \alpha(s, T_i) &= \sum_{i=0}^N w_i \cdot \left[a(T_i) \cdot [s - S(t)] + \frac{P(t, T_i)}{An(t)} \right] \\ &= \left[\sum_{i=0}^N w_i \cdot a(T_i) \right] \cdot [s - S(t)] + S(t). \end{aligned}$$

This yields the consistency condition as

$$\sum_{i=0}^N w_i \cdot a(T_i) = 1. \quad (10)$$

The previous equation shows that the basis spreads do have an impact on the slope of the annuity mappin function and thus affect the convexity adjustment.

The additivity and (multi-curve) consistency condition impose restrictions on the function $a(\cdot)$ on sets of discrete points in time. For CMS pricing $a(T_p)$ for $T_p \in [\bar{T}_0, \bar{T}_1]$ seems most important as CMS payment dates are often at the end of the accrual period following the fixing. To actually arrive at a specific model a parametric form of the slope function $a(\cdot)$ needs to be specified. This choice obviously defines the quality and flexibility of the linear TSR model. Fairly sophisticated choices for $a(\cdot)$ are presented e.g. in [AP10, 16.3.2] and [CP12]. To comply with our basis spread model in Sec. 1.2 these methods would need to be adapted to the multi-curve consistency condition (10).

Affine parameter functions. Here we want to focus on the impact of the basis. Thus we choose the functional form for $a(\cdot)$ as simple as possible. We consider the affine structures

$$a(T_p) = u \cdot (T_N - T_p) + v. \quad (11)$$

Then additivity and consistency condition become

$$\sum_{j=1}^M \bar{\tau}_j \cdot [u \cdot (T_N - \bar{T}_j) + v] = u \cdot \left[\sum_{j=1}^M \bar{\tau}_j \cdot (T_N - \bar{T}_j) \right] + v \cdot \left[\sum_{j=1}^M \bar{\tau}_j \right] = 0$$

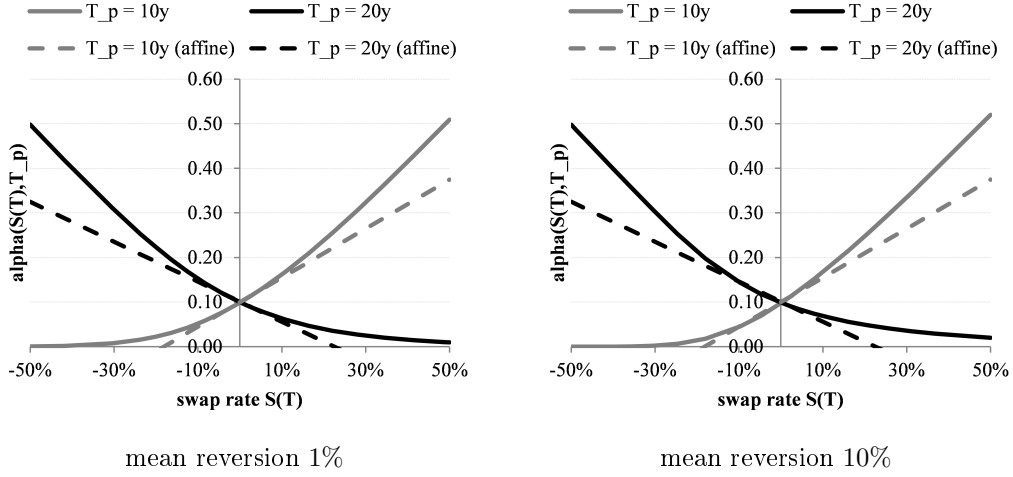


Figure 3: Annuity mapping function $\alpha(\cdot, T_p)$ for a 10y x 10y swap rate in the linear TSR model with affine parameter function (dashed lines) and in a Hull White model ($T_p \in \{10y, 20y\}$) calibrated to EUR market data as of September 2015 (mean reversion 1% and 10%).

and

$$\sum_{i=0}^N w_i \cdot [u \cdot (T_N - T_i) + v] = u \cdot \left[\sum_{i=0}^N w_i \cdot (T_N - T_i) \right] + v \cdot \left[\sum_{i=0}^N w_i \right] = 1.$$

Note that $T_N = \bar{T}_M$ and thus the last term in the above sums after u vanishes. The linear equations may be solved as

$$u = - \frac{\left[\sum_{j=1}^M \bar{\tau}_j \right]}{\left[\sum_{j=1}^{M-1} \bar{\tau}_j \cdot (T_N - \bar{T}_j) \right] \cdot \left[\sum_{i=0}^N w_i \right] - \left[\sum_{i=0}^{N-1} w_i \cdot (T_N - T_i) \right] \cdot \left[\sum_{j=1}^M \bar{\tau}_j \right]} \quad (12)$$

and

$$v = \frac{\left[\sum_{j=1}^{M-1} \bar{\tau}_j \cdot (T_N - \bar{T}_j) \right]}{\left[\sum_{j=1}^{M-1} \bar{\tau}_j \cdot (T_N - \bar{T}_j) \right] \cdot \left[\sum_{i=0}^N w_i \right] - \left[\sum_{i=0}^{N-1} w_i \cdot (T_N - T_i) \right] \cdot \left[\sum_{j=1}^M \bar{\tau}_j \right]}. \quad (13)$$

As a result we find that the annuity mapping function is fully specified by the schedule dates (with corresponding year fractions) and the basis spreads encoded in the weights w_i given in (3).

We compare the resulting annuity mapping function of the linear TSR model with affine parameter function to the Hull White model implied annuity mapping function in Figure 3 and Figure 4.

3 Application to Non-Standard Libor and CMS Coupons

Given the annuity mapping function representation in (9) we get for a CMS payoff

$$\begin{aligned} V(t) &= An(t) \cdot E^{An} \left[\left[a(T_p) \cdot [S(T) - S(t)] + \frac{P(t, T_p)}{An(t)} \right] \cdot h(S(T)) \right] \\ &= P(t, T_p) \cdot \left\{ E^{An} [h(S(T))] + a(T_p) \cdot \frac{An(t)}{P(t, T_p)} \cdot E^{An} [[S(T) - S(t)] \cdot h(S(T))] \right\}. \end{aligned} \quad (14)$$

Equation (14) is a general representation of the pricing formulas for CMS caps, floors and swaps in [Hag03, Eq. (3.4 a - c)]. Our representation only differs in the specification of the slope function $a(\cdot)$ given by (11) with (12) and (13).

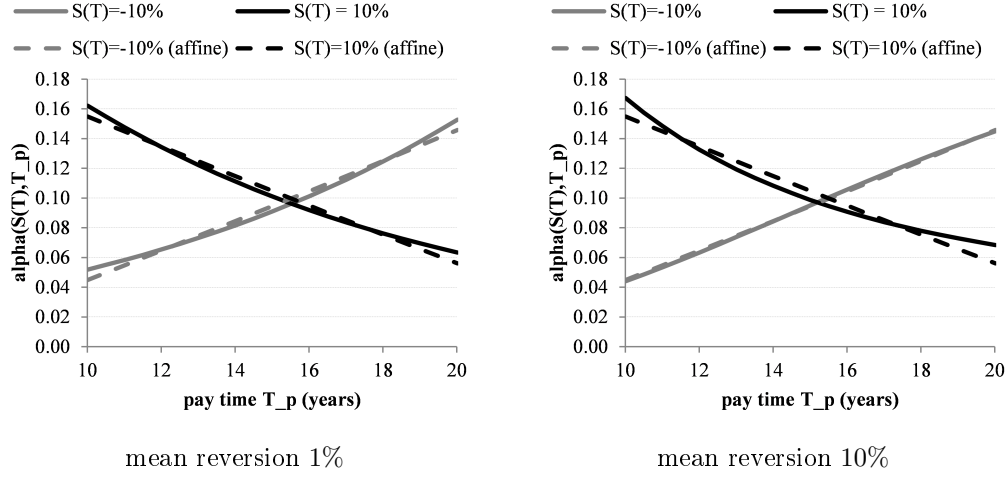


Figure 4: Annuity mapping function $\alpha(S(T), \cdot)$ for a 10y x 10y swap rate in the linear TSR model with affine parameter function (dashed lines) and in a Hull White model ($S(T) \in \{-10\%, 10\%\}$) calibrated to EUR market data as of September 2015 (mean reversion 1% and 10%).

Using the representation in (14) we get the CMS convexity adjustment as

$$CA(t) = a(T_p) \cdot \frac{An(t)}{P(t, T_p)} \cdot E^{An} [[S(T) - S(t)] \cdot h(S(T))].$$

We find that tenor basis only affects the factor $a(T_p)$. In particular the evaluation of the expectation $E^{An} [[S(T) - S(t)] \cdot h(S(T))]$ is not affected by the tenor basis model. This expectation can be evaluated by full replication or approximated by log-normal or normal models. For caps, floors and swaps log-normal model approximations are given e.g. in [Hag03, Eq.'s (3.5)]. Since normal model representations are of particular interest in current low interest rate market environment we derive the corresponding CMS formulas in Appendix A.1.

In the following subsections we analyse the impact of the basis on $a(T_p)$.

3.1 Libor Adjustments

In the previous section we elaborated on models for a general forward swap rate $S(t)$. The analysis also specialises to forward Libor rates. Note that forward Libor rates may be considered single-period swap rates with equal year fractions for the fixed and floating leg. For a Libor rate spanning from T_0 to T_1 we have

$$L(t) = L(t; T_0, T_1) = \frac{D(T_0, T_1) \cdot P(t, T_0) - P(t, T_1)}{\tau_1 \cdot P(t, T_1)}.$$

We measure time differences $T_1 - T_0$ in year fractions $\tau(T_0, T_1) = \tau_1$ and get

$$u = \frac{1}{D(T_0, T_1) \cdot \tau_1} \text{ and } v = 0.$$

Consequently, the annuity mapping slope function becomes $a(T_p) = \tau(T_p, T_1) / [D(T_0, T_1) \cdot \tau_1]$ which yields the convexity adjustment

$$CA(t) = \frac{\tau(T_p, T_1)}{D(T_0, T_1)} \cdot \frac{P(t, T_1)}{P(t, T_p)} \cdot E^{An} [[L(T) - L(t)] \cdot h(L(T))].$$

The impact of the tenor spread may be estimated by approximating $D(T_0, T_1)^{-1} \approx 1 - \int_{T_0}^{T_1} b(\cdot, u) du$. Then

$$CA(t) = \left[1 - \int_{T_0}^{T_1} b(\cdot, u) du \right] \cdot \tau(T_p, T_1) \cdot \frac{P(t, T_1)}{P(t, T_p)} \cdot E^{An} [[L(T) - L(t)] \cdot h(L(T))].$$

That is, given our deterministic spread assumption, we find that tenor basis has a small relative impact on the overall convexity adjustment. The relative impact is proportional to the overall basis spread size and Libor period. For positive tenor spreads (which is usually the case for Libor versus OIS) tenor basis slightly reduces the convexity adjustment.

Libor in-arrears. For the special case of Libor in-arrears with $T_p = T_0$ we get

$$\begin{aligned} CA(t) &= \frac{\tau(T_0, T_1)}{D(T_0, T_1)} \cdot \frac{P(t, T_1)}{P(t, T_0)} \cdot E^{An} [[L(T) - L(t)] \cdot h(L(T))] \\ &= \frac{\tau(T_0, T_1)}{1 + \tau_1 \cdot L(t)} \cdot E^{An} [[L(T) - L(t)] \cdot h(L(T))]. \end{aligned}$$

Note that this result could also be derived only by the the general definition of the convexity adjustment in (4) and our deterministic spread assumption in (1). In particular the result is independent of a particular annuity mapping function. As a result we find that tenor basis only enters in-arrears convexity adjustment by estimating $L(t)$ in a multi-curve framework. Note that this result relies on the assumption of deterministic *continuous compounded* tenor spreads.

In a single-curve framework the term $1 + \tau_1 \cdot L(t)$ could be replaced by $P(t, T_0)/P(t, T_1)$. Obviously, this is in general not correct in our tenor spread setting. However, the replacement turns out to be correct (and necessary) if we define the tenor basis in terms of deterministic *simple compounded* spreads. For details see Appendix A.4.

3.2 CMS Adjustments

We mentioned before that the tenor basis enters the CMS convexity adjustment by the slope function $a(T_p)$. To analyse the qualitative impact on general CMS payoffs we apply some simplifications:

- Assume equal fixed and floating leg schedules with fixed year fraction τ
- Assume flat linearised spreads $\tau \cdot b = D_i - 1$

With these simplifications the slope function $a(T_p)$ becomes

$$a(T_p) \approx \frac{T_N - T_p - \tau \cdot (N - 1)/2}{N \cdot \tau} \cdot (1 - b \cdot \tau).$$

For details of the derivation see Appendix A.6.

As a result we find that the tenor basis has a small relative impact on general CMS convexity adjustments. Analogously as for the Libor adjustment the relative impact is proportional to the overall basis spread size and Libor period. And again for positive tenor spreads (which is usually the case for Libor versus OIS) tenor basis slightly reduces the convexity adjustment.

4 Conclusion

We analysed the impact of individual curves for forward rate and discount factor evaluation on the construction of an annuity mapping function for CMS pricing. It turns out that the basis between forward and discount curve impacts the consistency condition which needs to be imposed on the annuity mapping function. Moreover we quantified the impact of the basis on the annuity mapping

function. As a result we find that the effect on the annuity mapping function is a relative reduction of the convexity adjustment in the order of the basis spreads. Thus the impact may be considered small compared to the impact of the basis on the underlying Vanilla pricing and the general uncertainty in choosing a particular annuity mapping function.

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A Appendix

A.1 Normal Model for CMS Caps, Floors and Swaps

To price CMS swaps we need to evaluate

$$E^{An} [[S(T) - S(t)] \cdot S(T)] = E^{An} [S(T) - S(t)]^2 = Var^{An} [S(T)].$$

A normal model approximation of the variance is given by

$$Var^{An} [S(T)] \approx [\sigma_n^{ATM}]^2 (T - t).$$

Here σ_n^{ATM} is the implied ATM normal volatility of the underlying rate.

For CMS caps ($\psi = +1$) and floors ($\psi = -1$) with strike rate K we need to evaluate

$$E^{An} [S(T) - S(t)] \cdot [\psi (S(T) - K)]^+.$$

Re-arranging terms yields

$$\begin{aligned} [S(T) - S(t)] \cdot [\psi (S(T) - K)]^+ &= [(S(T) - K) - (S(t) - K)] \cdot [\psi (S(T) - K)]^+ \\ &= (S(T) - K) \cdot [\psi (S(T) - K)]^+ - (S(t) - K) \cdot [\psi (S(T) - K)]^+ \\ &= \psi \cdot 1_{\{\psi(S(T)-K) \geq 0\}} \cdot [S(T) - K]^2 - [S(t) - K] \cdot [\psi (S(T) - K)]^+ \end{aligned}$$

Hence we need to evaluate a Vanilla option and a power option given by

$$E^{An} [\psi (S(T) - K)]^+ \quad \text{and} \quad E^{An} [1_{\{\psi(S(T)-K) \geq 0\}} [S(T) - K]^2].$$

The Vanilla option can be evaluated *exactly* using the implied normal volatility σ_n^K for the strike K and Bachelier's formula

$$E^{An} \left[[\psi(S(T) - K)]^+ \right] = \psi(S(t) - K) \Phi \left(\frac{\psi(S(t) - K)}{\sigma_n^K \sqrt{T-t}} \right) + \phi \left(\frac{\psi(S(t) - K)}{\sigma_n^K \sqrt{T-t}} \right) \sigma_n^K \sqrt{T-t}.$$

Here Φ denotes the cummulated standard normal distribution function and $\phi(x) = \Phi'(x)$. Bachelier's formula is based on the assumption that $S(T)$ is normally distributed with mean $S(t)$ and standard deviation $\nu = \sigma_n^K \sqrt{T-t}$. Re-using these assumptions we may evaluate the expectation of the power option as

$$E^{An} \left[1_{\{\psi(S(T)-K) \geq 0\}} [S(T) - K]^2 \right] = \int_{\psi(s-K) \geq 0} [s - K]^2 \phi \left(\frac{s - S(t)}{\nu} \right) \frac{1}{\nu} ds.$$

Substituting $u = (s - S(t)) / \nu$ and $h = (S(t) - K) / \nu$ yields

$$\begin{aligned} E^{An} \left[1_{\{\psi(S(T)-K) \geq 0\}} [S(T) - K]^2 \right] &= \int_{\psi(\nu u + S(t) - K) \geq 0} [\nu u + S(t) - K]^2 \phi(u) du \\ &= \int_{\psi(u+h) \geq 0} \nu^2 [u + h]^2 \phi(u) du. \end{aligned}$$

It turns out that

$$\int [u + h]^2 \phi(u) du = (h^2 + 1) \Phi(u) - (u + 2h) \phi(u).$$

Thus we get for call options

$$\begin{aligned} E^{An} \left[1_{\{\psi(S(T)-K) \geq 0\}} [S(T) - K]^2 \right] &= \nu^2 [(h^2 + 1) \Phi(u) - (u + 2h) \phi(u)]_{u=-h}^{+\infty} \\ &= \nu^2 [(h^2 + 1) (1 - \Phi(-h)) + (2h - h) \phi(-h)] \\ &= \nu^2 [(h^2 + 1) \Phi(h) + h \phi(h)]. \end{aligned}$$

Analogously put options become

$$\begin{aligned} E^{An} \left[1_{\{(K-S(T)) \geq 0\}} [S(T) - K]^2 \right] &= \nu^2 [(h^2 + 1) \Phi(u) - (u + 2h) \phi(u)]_{u=-\infty}^{-h} \\ &= \nu^2 [(h^2 + 1) \Phi(-h) - (2h - h) \phi(-h)] \\ &= \nu^2 [(h^2 + 1) \Phi(-h) - h \phi(-h)]. \end{aligned}$$

Combining put and call options yields

$$E^{An} \left[1_{\{\psi(S(T)-K) \geq 0\}} [S(T) - K]^2 \right] = \nu^2 [(h^2 + 1) \Phi(\psi h) + \psi h \phi(\psi h)].$$

Finally we get

$$\begin{aligned} E^{An} \left[[S(T) - S(t)] \cdot [\psi(S(T) - K)]^+ \right] &= \psi \nu^2 [(h^2 + 1) \Phi(\psi h) + \psi h \phi(\psi h)] \\ &\quad - \nu h [\psi \nu h \Phi(\psi h) + \phi(\psi h) \nu] \\ &= (\psi \nu^2 h^2 + \psi \nu^2 - \psi \nu^2 h^2) \Phi(\nu h) + (\nu^2 h - \nu^2 h) \phi(\psi h) \\ &= \psi \nu^2 \Phi(\nu h). \end{aligned}$$

As a result we conclude that the convexity adjustment for CMS caps and floors is approximated by

$$E^{An} \left[[S(T) - S(t)] \cdot [\psi(S(T) - K)]^+ \right] \approx \psi [\sigma_n^K]^2 (T-t) \Phi \left(\psi \frac{S(t) - K}{\sigma_n^K \sqrt{T-t}} \right) \quad (15)$$

with σ_n^K beeing the implied normal volatility for the strike rate K .¹

¹Formula (15) is also stated in Eq.'s (3.6c,d) of a preprint version of [Hag03].

A.2 Alternative linear TSR model for Libor payoffs.

For the special case that the coupon is paid at the period start date (e.g. in-arrears payments) the annuity mapping function may be written as

$$\alpha(l, T_0) = \frac{1}{\tau_1} \cdot \frac{1 + \tau_1 \cdot l}{1 + \tau_1 \cdot L(t)} \cdot \frac{P(t, T_0)}{P(t, T_1)}.$$

This formula for the in-arrears payments inspires the following alternative formulation of the annuity mapping function

$$\alpha(l, T_p) = \frac{1}{\tau_1} \cdot \frac{1 + \tau(T_p, T_1) \cdot l}{1 + \tau(T_p, T_1) \cdot L(t)} \cdot \frac{P(t, T_p)}{P(t, T_1)}.$$

This formulation also satisfies the no-arbitrage, additivity and consistency condition. However, it differs from the approach above in that the function $a(\cdot)$ is not affine in T_p anymore. As a consequence the formula does not easily generalise to swap rates.

A.3 CMS Adjustment for Simple Compounded Spreads

Alternatively to the spread model in Sec. 1.2 we could assume a deterministic spread between (forward) Libor rates and forward rates projected from the discount curve (e.g. *OIS forwards*). That is we consider

$$L(t; T_{i-1}, T_i) = \left[\frac{P(t, T_{i-1})}{P(t, T_i)} - 1 \right] \frac{1}{\tau_i} + B(T_{i-1}, T_i). \quad (16)$$

Here $B(T_{i-1}, T_i)$ represents the Libor versus discounting spread. Substituting the previous relation into the definition of the swap rate yields

$$S(t) = \frac{P(t, T_0) - P(t, T_N) + \sum_{i=1}^N B(T_{i-1}, T_i) \cdot \tau_i \cdot P(t, T_i)}{An(t)}.$$

Thus we can come up again with the swap rate representation (2) as

$$S(t) = \frac{\sum_{i=0}^N w_i P(t, T_i)}{An(t)}.$$

but with alternative weights w_i as

$$\begin{aligned} w_0 &= 1, \\ w_i &= B(T_{i-1}, T_i) \cdot \tau_i \text{ for } i = 1, \dots, N-1 \text{ and} \\ w_N &= -1 + B(T_{N-1}, T_N) \cdot \tau_N. \end{aligned}$$

The analysis in Sec. 2 remains unchanged.

A.4 Libor In-arrears Adjustment for Simple Compounded Spreads

The general convexity adjustment for in-arrears payoffs with $T_p = T_0$ is

$$CA(t) = E^{An} \left[\left[\frac{P(t, T_1)}{P(t, T_0)} \cdot \frac{P(T, T_0)}{P(T, T_1)} - 1 \right] \cdot h(S(T)) \right].$$

Re-arranging terms in (16) yields

$$\frac{P(t, T_0)}{P(t, T_1)} = 1 + \tau_1 L(t; T_0, T_1) - \tau_1 B(T_0, T_1).$$

Applying this to the convexity adjustment yields

$$\begin{aligned} \frac{P(t, T_1)}{P(t, T_0)} \cdot \frac{P(T, T_0)}{P(T, T_1)} - 1 &= \frac{1 + \tau_1 L(T) - \tau_1 B(T_0, T_1)}{1 + \tau_1 L(t) - \tau_1 B(T_0, T_1)} - 1 \\ &= \frac{\tau_1 [L(T) - L(t)]}{1 + \tau_1 L(t) - \tau_1 B(T_0, T_1)} \end{aligned}$$

and thus

$$\begin{aligned} CA(t) &= \frac{\tau_1}{1 + \tau_1 L(t) - \tau_1 B(T_0, T_1)} \cdot E^{An} [[L(T) - L(t)] \cdot h(S(T))] \\ &= \tau_1 \frac{P(t, T_1)}{P(t, T_0)} \cdot E^{An} [[L(T) - L(t)] \cdot h(S(T))]. \end{aligned}$$

As a result we find that the convexity adjustment does depend on the choice of tenor spread model and the variance between both approaches is in the order of the the spread itself.

A.5 Single Curve Approximation to Multi-Curve CMS Pricing

In the case that a pricing system may not be able to handle separate discount and forward curves we might resort to the swap rate spread representation in [CP12]. That is we consider

$$S(t) = \underbrace{\frac{P(t, T_0) - P(t, T_N)}{An(t)}}_{\bar{S}(t)} + B_s(T_0, T_N)$$

for a deterministic swap spread $B_s(T_0, T_N)$. The quantity $\bar{S}(t)$ now represents the corresponding swap rate in a single-curve setting using only the discount curve. In that case we can write the CMS payoff as

$$V(t) = An(t) \cdot E^{An} [\alpha (\bar{S}(T) + B_s(T_0, T_N), T_p) \cdot h(\bar{S}(T) + B_s(T_0, T_N))] .$$

It turns out that if we have a linear TSR model for $S(T)$ then the corresponding model for $\bar{S}(T)$ is also a linear TSR model. However, the consistency condition (10) simplifies to

$$a(T_0) - a(T_N) = 1.$$

Besides the annuity mapping function the expectation for the convexity adjustment needs to be adapted. For caps and floors it becomes

$$E^{An} [S(T) - S(t)] \cdot [\psi(S(T) - K)]^+ = E^{An} [\bar{S}(T) - \bar{S}(t)] \cdot [\psi(\bar{S}(T) - [K - B_s(T_0, T_N)])]^+ .$$

Thus the swap rate spread translates into a shift in the option strike.

A.6 CMS Annuity Mapping Function Approximations

In order to get a better understanding of the qualitative impact of the basis on the annuity mapping function for CMS payoffs we approximate the terms in (12) and (13) by

$$\begin{aligned} \sum_{j=1}^M \bar{\tau}_j &\approx \sum_{i=1}^N \tau_i \approx N \cdot \tau \\ \sum_{j=1}^{M-1} \bar{\tau}_j \cdot (T_N - \bar{T}_j) &\approx \sum_{i=1}^N \tau_i \cdot (T_N - T_i) \approx N \cdot \frac{N-1}{2} \cdot \tau^2, \end{aligned}$$

$$\begin{aligned}
\sum_{i=0}^N w_i &= \sum_{i=1}^N D_i - 1 \approx b \cdot N \cdot \tau, \text{ and} \\
\sum_{i=0}^{N-1} w_i \cdot (T_N - T_i) &= (T_N - T_0) + \sum_{i=0}^{N-1} (D_{i+1} - 1) \cdot (T_N - T_i) \\
&\approx (T_N - T_0) + b \cdot (T_N - T_0) \cdot \frac{(T_N - T_0) + (T_N - T_{N-1})}{2} \\
&= (T_N - T_0) \cdot \left[1 + b \cdot \frac{(T_N - T_0) + (T_N - T_{N-1})}{2} \right] \\
&\approx N \cdot \tau \left[1 + b \frac{(N+1) \cdot \tau}{2} \right]
\end{aligned}$$

Here b is considered the flat or average basis spread. Furthermore, we assume $M = N$ and both schedules with constant year fraction τ . The denominator for u and v becomes

$$-N^2 \cdot \tau^2 \cdot \left[1 + b \frac{N+1 - (N-1)}{2} \cdot \tau \right] = -N^2 \cdot [1 + b \cdot \tau].$$

Finally, the slope function parameters become

$$u = \frac{1}{N \cdot \tau \cdot [1 + b \cdot \tau]} \text{ and } v = -\frac{(N-1)/2}{N \cdot [1 + b \cdot \tau]}.$$

The slope function itself is thus

$$a(T_p) = \frac{T_N - T_p - \tau \cdot (N-1)/2}{N \cdot \tau \cdot [1 + b \cdot \tau]} \approx \frac{T_N - T_p - \tau \cdot (N-1)/2}{N \cdot \tau} \cdot (1 - b \cdot \tau).$$